

SQL Assignment

End-to-End Database Management Use Case

Objective

The goal of this assignment is to understand and apply the full spectrum of SQL capabilities by designing, managing, and querying a relational database. Students will work through a practical business scenario and use **DDL, DML, DQL, TCL, and DCL commands** to build and operate the database.

This assignment evaluates the ability to:

- Design a relational schema
- Insert and manage data
- Query and analyze information
- Manage transactions safely
- Control access to database objects

Business Scenario

A company wants to build a **Retail Sales Management System** to track customers, products, orders, and employees. The database should allow the organization to store transactional data, analyze sales, and control user access.

You are hired as a **Database Analyst** to design and manage the database system using SQL.

Your task is to implement the system using different SQL language categories.

Assignment Requirements

You must complete the assignment using the following SQL language groups.

1. Data Definition Language (DDL)

Design the database structure.

Tasks include:

- Creating tables for core business entities such as:
 - Customers
 - Products
 - Orders
 - Order Items
 - Employees
- Defining **primary keys and foreign keys**
- Setting **constraints** such as NOT NULL and UNIQUE
- Creating **indexes** to improve query performance
- Modifying the schema using ALTER statements
- Dropping or restructuring tables when required

The schema should reflect a **normalized relational design** that maintains data integrity.

2. Data Manipulation Language (DML)

Populate and manage the data in the database.

Tasks include:

- Inserting sample records into all tables
- Updating existing data based on business rules
- Deleting specific records when required
- Performing bulk inserts where necessary

Ensure that inserted data maintains **referential integrity**.

3. Data Query Language (DQL)

Analyze and retrieve insights from the database.

Write SQL queries that demonstrate:

- Basic data retrieval using SELECT
- Filtering records using WHERE
- Sorting using ORDER BY
- Aggregations using COUNT, SUM, AVG, MAX, MIN
- GROUP BY and HAVING
- Joining multiple tables
- Subqueries
- Window functions such as ranking or running totals

Queries should reflect **real business analytics scenarios**.

4. Transaction Control Language (TCL)

Demonstrate transaction management to ensure data consistency.

Tasks include:

- Starting a transaction
- Creating savepoints
- Rolling back changes
- Committing successful transactions

Show how transactions prevent **data corruption during failures**.

5. Data Control Language (DCL)

Manage user permissions for database security.

Tasks include:

- Creating database users or roles
- Granting specific privileges on tables
- Revoking permissions when access is no longer required

Ensure proper **access control policies** for different users.

Expected Deliverables

Students must submit:

1. A **SQL script file** containing all commands used in the assignment.
2. Database schema creation scripts.
3. Data insertion scripts.

4. SQL queries used for analysis.
5. Transaction and permission management commands.

The script should be organized with clear sections for:

- DDL
 - DML
 - DQL
 - TCL
 - DCL
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Evaluation Criteria

Submissions will be evaluated based on:

- Correct database design
- Proper use of SQL syntax
- Use of constraints and relationships
- Query correctness and efficiency
- Demonstration of transactions and permissions
- Clarity and organization of the SQL script

ER Diagram is must